

Business Rules Baseline

Performance Boundary Review - Business Semantics Unchanged

Document Control	Value
Document ID	BR-RR-BASELINE
Version	1.6 - Performance Review Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Supersedes	BR-RR-BASELINE v1.5
Revision Scope	Document how performance optimization is constrained by locked business rules; no BR is added, removed or changed.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

กฎธุรกิจฉบับฐานกลาง / BUSINESS RULES BASELINE

Repair Request MVP - Baseline Revision

Document Control	Value
Document ID	BR-RR-BASELINE
Version	1.5 - Baseline Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	RR-REQ-001 v1.5; prior BR v1.4; D-01..D-15
Audience	Business Owner, SA, Developer, Tester, Project Manager
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	รวมและลือกฎธุรกิจ BR-01 ถึง BR-19 รวม audit closure clarifications.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Definitions & Locked Decisions

Term	Definition
Customer Acceptance Contact	Exactly one designated Requester or active Site Contact in the Work Order Site; only this contact may Accept/Reject.
Completed	Customer Acceptance Contact accepted the work; operational work done but not yet Supervisor-closed.
Closed	Supervisor verified Work Summary and Cost Summary and closed/locked the Work Order.
Converted	Repair Request after one Work Order is created; Requester can no longer edit/cancel.
Corrective Action	Created automatically from Customer Reject; Team Lead plan -> Supervisor approval -> rework -> review -> same contact re-acceptance.
Resolution SLA	24h 24/7 from SUBMITTED; Risk at T-4h; stop only defined terminal states; breach sticky.
Active Request for BR-14	SUBMITTED/UNDER_REVIEW/APPROVED plus CONVERTED while linked WO not CLOSED/CANCELLED.

ID	Topic	Decision
D-11	Identity / Customer Mapping	Requester = authenticated user that owns the Draft (mapped to created_by). Customer/Company security scope = tenant_id . Site = site_id . Client must not supply tenant/actor as authority.
D-12	Duplicate Active Set	BR-14 "active Request" = SUBMITTED, UNDER_REVIEW, APPROVED, and CONVERTED only while its linked Work Order is not CLOSED/CANCELLED. DRAFT, REJECTED and CANCELLED are excluded.
D-13	SLA Risk Threshold	MVP "Near Breach / SLA Risk" is reached at 4 hours before effective_due_at . When an authorized SLA Override changes effective_due_at, risk_at is recalculated; an existing breached_at is never cleared.
D-14	Notification Routing	Every baseline notification has deterministic recipient(s) defined by the Notification Recipient Matrix. Delivery failure is retryable and must not roll back the business transition.
D-15	MISSED Decision Semantics	Original MISSED Service Visit remains immutable. Reschedule/Follow-up/Reassign creates a new SCHEDULED visit referencing the MISSED visit. Cancel decision records "no follow-up"; no new visit is created and the Work Order continues unless separately cancelled.

2. Locked Business Rules BR-01..BR-19

Rule ID	Trigger / Actor	Condition / Guard	System Result / Audit
BR-01	Submit / Requester	Request No unique; identity immutable	Generate immutable Request No; audit
BR-02	Any significant action / all roles	Server-side mandatory/master/role/scope/state/concurrency validation	Reject invalid action; no state change/partial write
BR-03	Convert / Coordinator	Request APPROVED; no existing WO; current version	Create exactly one WO OPEN + Request CONVERTED atomically
BR-04	Reject/Cancel/Reassign/Reschedule/Override/Correction	Reason required	Persist reason, actor, time, state or old/new
BR-05		Assigned SCHEDULED Visit; no active overlap	Session CHECKED_IN; Visit/WO IN_PROGRESS

Rule ID	Trigger / Actor	Condition / Guard	System Result / Audit
	Check-in / Assigned Technician		
BR-06	Check-out / Assigned Technician	Summary+Outcome + required CLEAN evidence; valid time sequence	Session/Visit completed; audit
BR-07	Accept/Reject / designated Contact	WO AWAITING_CUSTOMER_ACCEPTANCE; contact active/same Site	Accept -> COMPLETED; Reject -> Corrective Action + reason
BR-08	Close / Supervisor	WO COMPLETED; Work Summary + Cost Summary reviewed	WO CLOSED; SLA stop; operation locked
BR-09	Cancel / Requester, Coordinator, Supervisor	Request pre-convert; Visit pre-check-in; WO pre-accept	Terminal cancellation + reason/audit; SLA stop only where defined
BR-10	MISSED / Coordinator	No Check-in	Original remains MISSED; no auto follow-up
BR-11	Time Correction	Supported timestamp only; valid sequence	Pending request; Supervisor decision; old/new/ reason stored
BR-12	SLA Start/Stop / System	Start SUBMITTED; stop Request REJECTED/ CANCELLED or WO CLOSED/CANCELLED	24h 24/7; sticky breach
BR-13	Correction/Override	Authorized; reason/sequence valid	Time Correction no SLA recalc; Override updates effective due only
BR-14	Submit duplicate check	Same Site + Eq/Location + Category <24h against active set	Warn; continuation reason required to proceed
BR-15	Customer Reject	Reason + designated contact	Create corrective cycle; same contact re-accepts
BR-16	Identity/Scope mapping	Authenticated context	created_by=Requester; tenant_id=Customer/ Company scope; never client authority
BR-17	SLA Risk	Current time >= effective_due_at - 4h and SLA active	Set/emit risk event; override recalculates future risk_at only
BR-18	Notification	Baseline business event committed	Create outbox event and deterministic recipients; failure retry independent
BR-19	MISSED decision	Original MISSED	Reschedule/Follow-up/Reassign creates new linked SCHEDULED visit; Cancel creates none; WO unchanged

3. Notification Recipient Matrix

Code	Trigger	Recipients	Business Object	Failure behavior
NTF-SUBMITTED	Request Submitted	Assigned Approver(s)	Repair Request	Create in-app event after commit; routing failure remains SUBMITTED
NTF-RETURNED	Return for Correction	Requester	Repair Request	Reason shown; Request is DRAFT
NTF-APPROVED	Request Approved	Coordinator(s) in Site scope	Repair Request	Action required: Convert
NTF-REJECTED	Request Rejected	Requester	Repair Request	Terminal; SLA stopped
NTF-ASSIGN	Assign/Reassign Visit	Assigned Technician + Team Lead	Service Visit	Include schedule; no sensitive leak
NTF-MISSED	Visit Missed	Coordinator + Team Lead	Service Visit	Decision required
NTF-SUMMARY	Work Summary Submitted	Supervisor	Work Order	Review required
NTF-ACCEPT	Customer Accepted	Supervisor + Coordinator	Work Order	Close guard remains Cost/Review
NTF-REJECT-WORK	Customer Rejected	Team Lead + Supervisor + Coordinator	Work Order/ Corrective Action	Corrective plan required

Code	Trigger	Recipients	Business Object	Failure behavior
NTF-SLA-RISK	SLA Risk (T-4h)	Coordinator + Supervisor	SLA Record	No state change
NTF-SLA-BREACH	SLA Breach	Coordinator + Supervisor	SLA Record	breached_at set once
NTF-TIME	Time Correction Decision	Correction requester	Time Correction	Show approve/reject result
NTF-CANCEL	Cancel	Requester + current operational owner as applicable	Request/Visit/WO	Terminal/propagation effect summarized

4. Rule-to-Source / Test Traceability

Rule	Primary FR	State / Event	Primary Test
BR-01	FR-01/02	ST-RR-002	TC-RR-001/003
BR-02	FR-01..09	all allowed actions	TC-SEC-001/002
BR-03	FR-04	ST-RR-008	TC-RR-010/011
BR-04	FR-03/05/08	Reject/Cancel/Reassign/Reschedule/Override/Correction	TC-AUD-001
BR-05	FR-06	ST-WS-001	TC-WO-006
BR-06	FR-06/07	ST-WS-004	TC-WO-007
BR-07	FR-07	ST-WO-005/007	TC-WO-009/010
BR-08	FR-07	ST-WO-006	TC-WO-011/TC-CST-001
BR-09	FR-03/05/07	Cancel transitions	TC-RR-009/TC-WO-004/012
BR-10	FR-05	ST-SV-008/009	TC-WO-004/005
BR-11	FR-08	ST-TC-*	TC-TIME-001/002
BR-12	FR-08	EV-SLA-001/005	TC-SLA-001
BR-13	FR-08	EV-SLA-004	TC-SLA-002/TC-TIME-002
BR-14	FR-02	ST-RR-002	TC-RR-004
BR-15	FR-07	ST-WO-007..010	TC-WO-010
BR-16	FR-01/09	security controls	TC-SEC-001
BR-17	FR-08	EV-SLA-002	TC-SLA-003
BR-18	FR-08/09	notification events	TC-NTF-001/002
BR-19	FR-05	ST-SV-009	TC-WO-005

5. Baseline Readiness and Change Control

Business rule semantics are complete for design once this revision is formally approved. Any change to BR-01..BR-19 or D-01..D-15 requires Change Request and impact review across State/UC/Data/Test/ERD/DB/API/UI.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

6. Performance Review Boundary - Business Rules Unchanged

BR-01 through BR-19 and D-01 through D-15 are unchanged. Performance work is allowed only as a Technical Design optimization that preserves the exact business result, state, role, guard, SLA, permission, audit and notification-recipient semantics.

Rule Area	Performance/Technical Guard	Trace
Authorization / Scope	Optimization, cache or read model must never bypass role, tenant, site, ownership or assignment checks.	BR-02/16 -> Security tests + Architecture
State-changing command	Concurrency optimization must still validate current state + rowversion and commit audit/outbox atomically.	BR-03..13,15,19 -> State/ API/DB/Test
Notification	Delivery remains post-commit/retryable; faster delivery may not roll back or alter committed business state.	BR-18 -> Worker/Outbox tests
SLA Risk/Breach	Worker cadence/batch/concurrency are technical configuration only; 24h/T-4h/sticky breach semantics remain fixed.	BR-12/17 -> SLA Worker tests
Files/Evidence	Streaming/storage optimization cannot make PENDING/ FAILED evidence usable or weaken authorized download.	BR-06/15 + file controls
Duplicate warning	Query/index optimization must preserve D-12 active set and BR-14 matching semantics exactly.	BR-14 -> SQL/EF query verification

7. Performance Decision Ownership

Latency percentile, load-test duration, worker batch size, page-size limit, caching and export execution strategy are **not Business Rules** unless formally approved as a requirement. They belong to Requirement NFR clarification, Architecture, API/DB/UI design and RR-PERF-001. Any technical choice that changes BR outcome requires Change Request.